

Project 1 – Network Security

**Create an Active Directory and promote it to a Domain Controller,
a Domain, Organizational Units (OUs), and User for each OU**

Introduction

This lab is designed to work on the Windows Server 2022 VM within the Azure Virtual Lab Environment. First part of the project is to install Active Directory (AD) and create Domain controller. Active Directory Domain Controller (AD DC) is a Windows Server running on Active Directory Domain Services (AD DS). AD DS acts as the central gatekeeper for a network, authenticating users, validating credentials, and enforcing security policies. Active Directory plays a vital role in cybersecurity because of its centralized control structure, it is a powerful administrative tool and a primary target for cyberattacks.

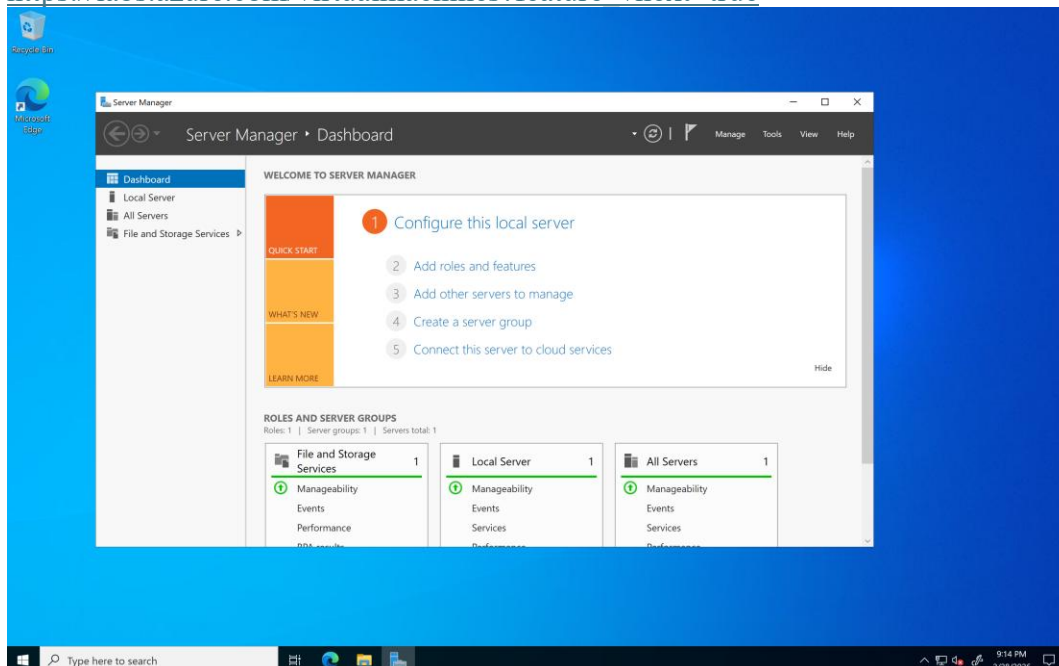
After installing AD and creating Domain Controller (DC), the server was promoted to a Domain Controller, and a new forest and domain named *academia.local* were created. Under the domain three Organizational Units named CSISUsers, HEALTHUsers, and BUSINESSUsers were established to logically organize network resources. Under each organization, user accounts were created for future use.

Completing part 1 to 3 of this lab provides hands-on experience with domain creation, directory structure organization, and user management. As a cybersecurity analyst, it is required to know about Active Directory because misconfiguration of AD can lead to severe security vulnerabilities.

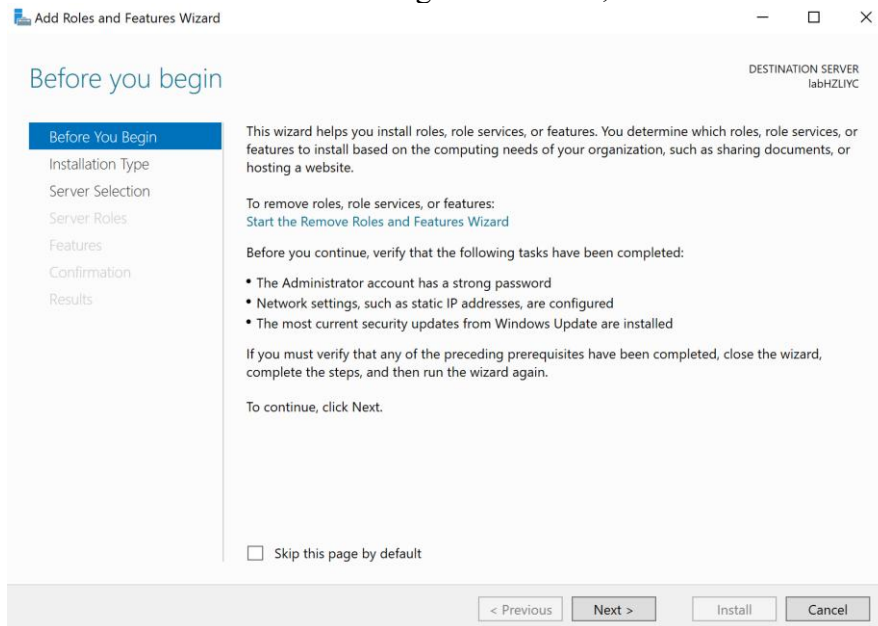
Part 1 - Installing Active Directory (AD) and Creating a Domain Controller

1. Logged to Azure Lab environment using following web link and logged into the Windows Server VM.

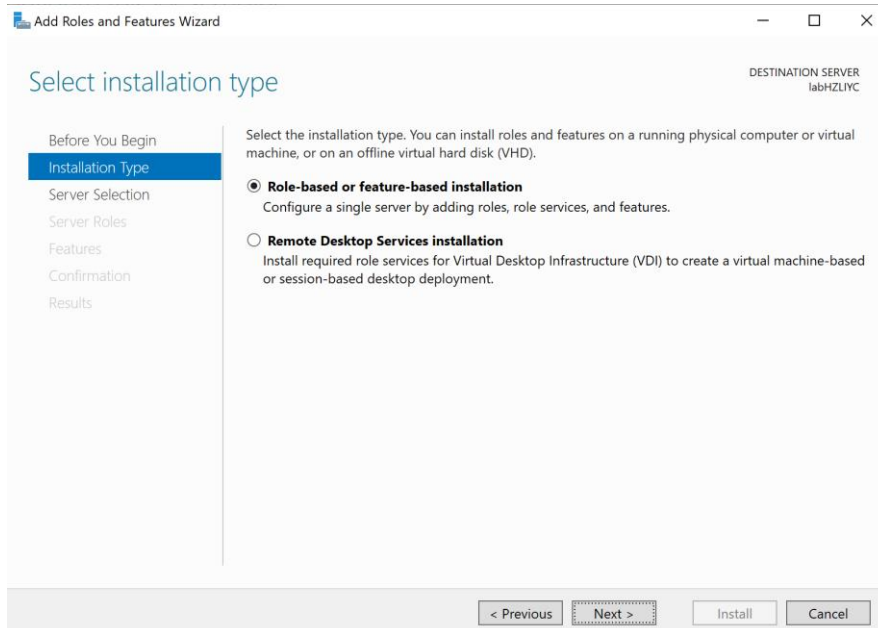
https://labs.azure.com/virtualmachines?feature_vnext=true



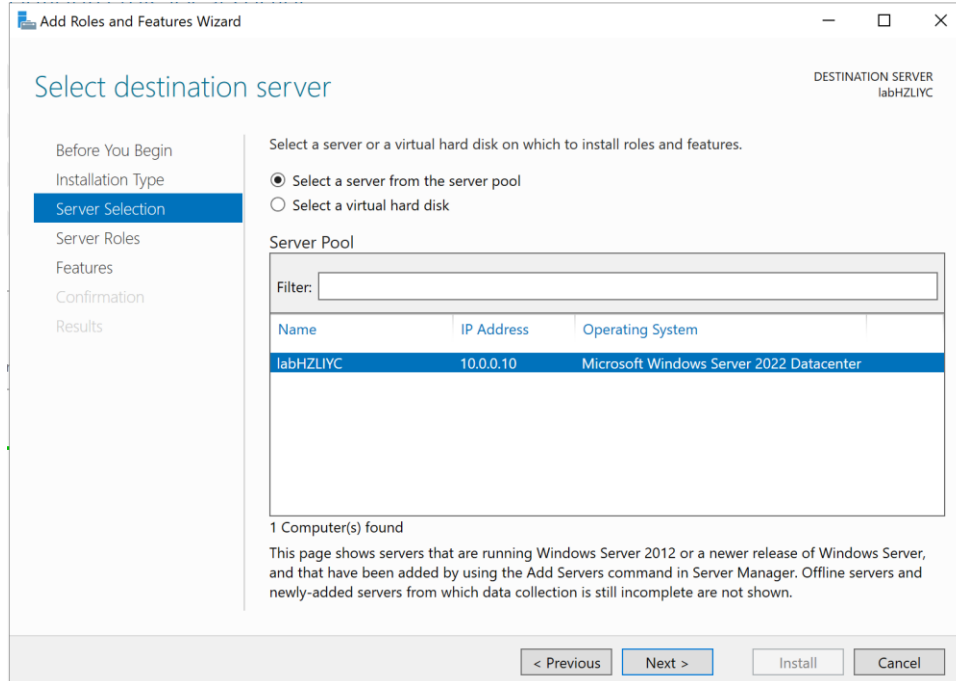
2. Within the 'Server Manager' dashboard, clicked 'Add Roles and Features'.



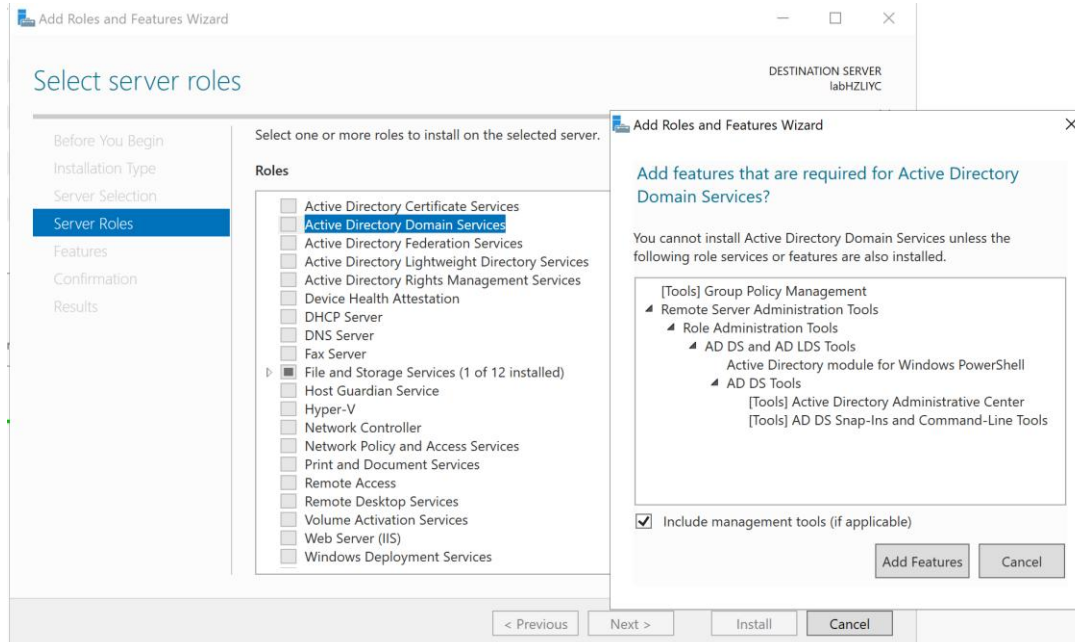
3. Selected 'Role-based or Feature-based installation' and clicked Next.



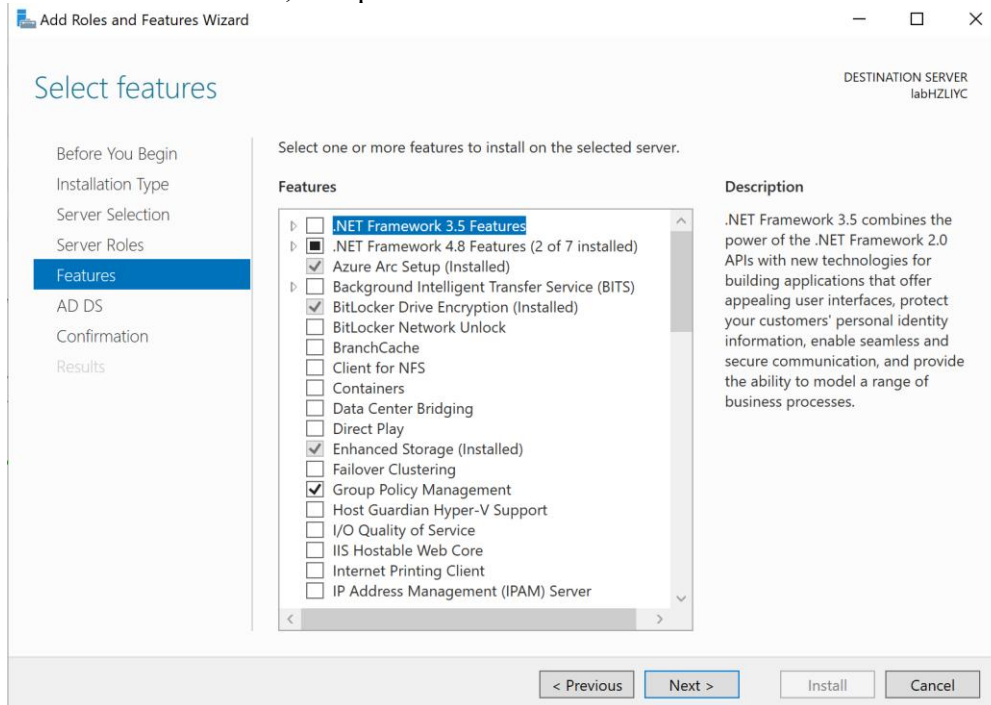
4. For 'Server Selection', chosen the option, 'Select a server from the server pool' and clicked Next.



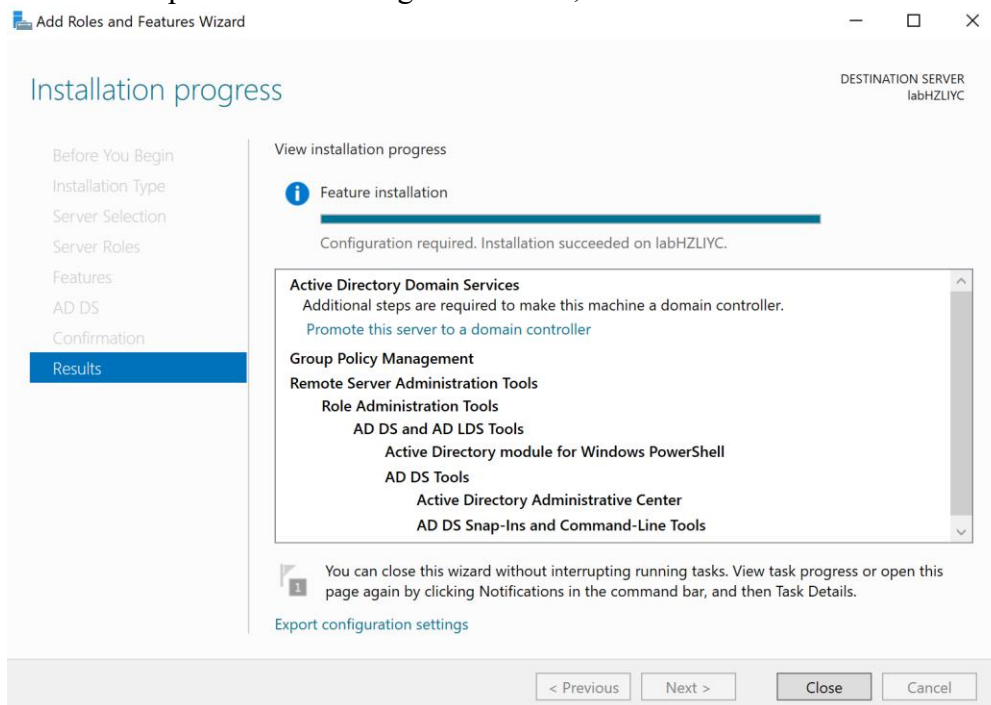
5. From the list of server roles, selected 'Active Directory Domain Services' and accepted to install the additional features. Then clicked Next.



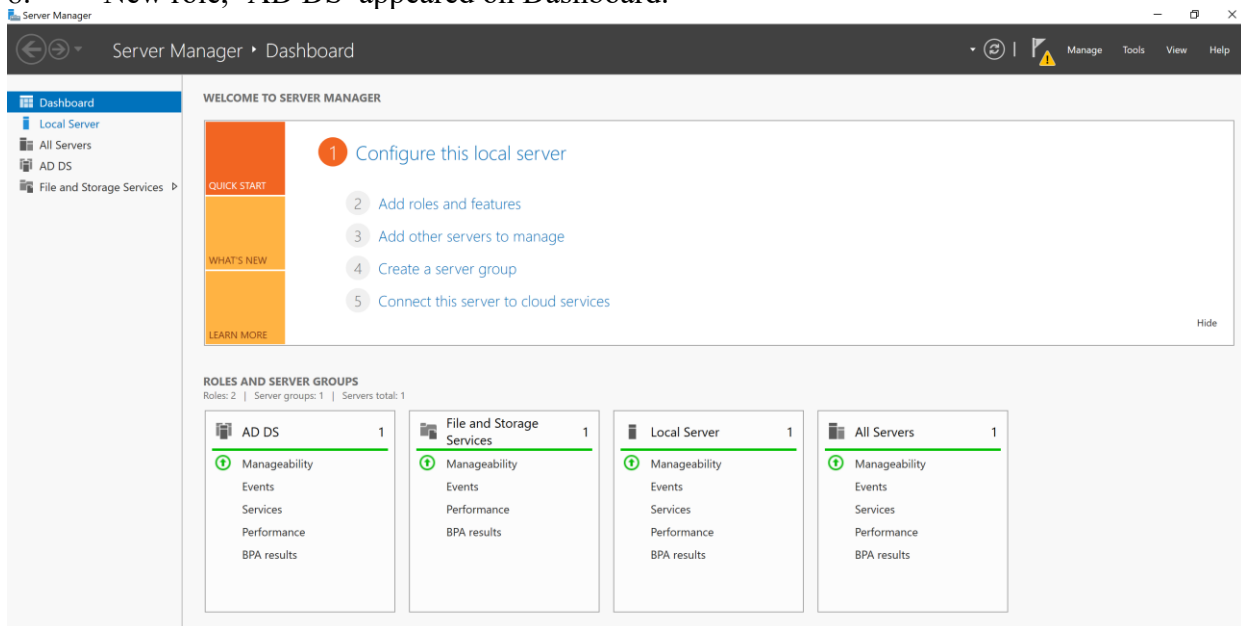
6. For 'Features', accepted default selection and clicked Next.



7. Accepted default settings for AD DS, Confirmation and clicked 'Install'.

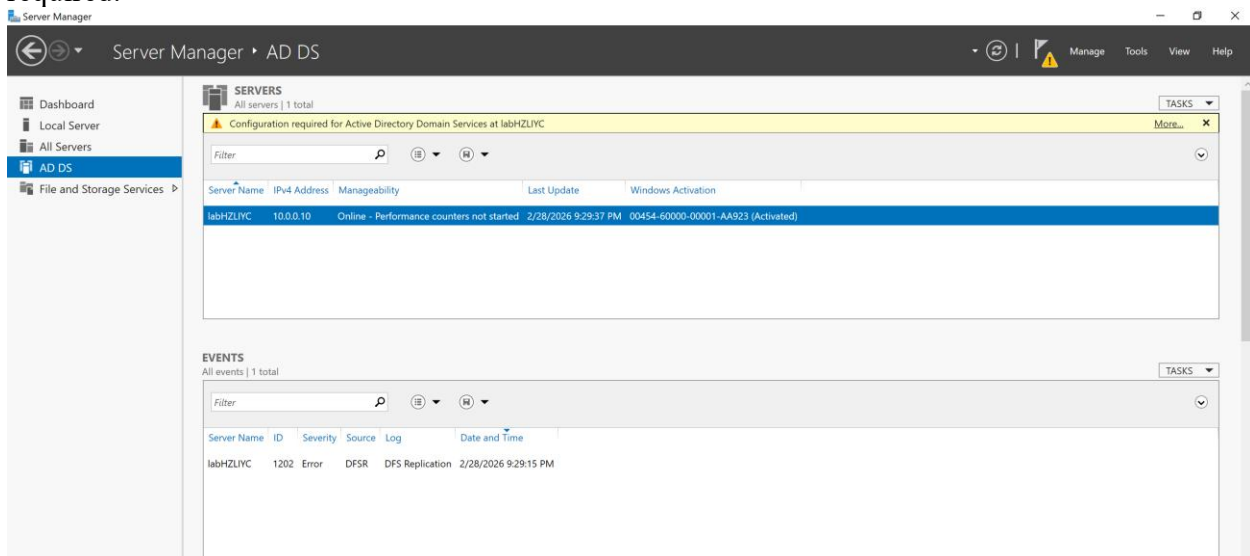


8. New role, 'AD DS' appeared on Dashboard.

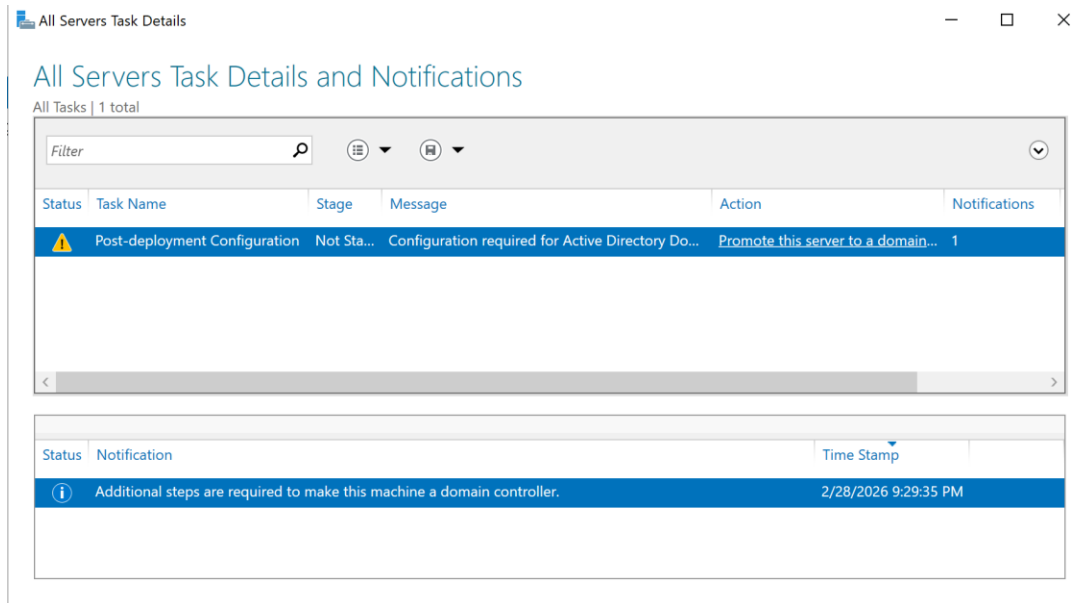


Promoting AD to a Domain Controller

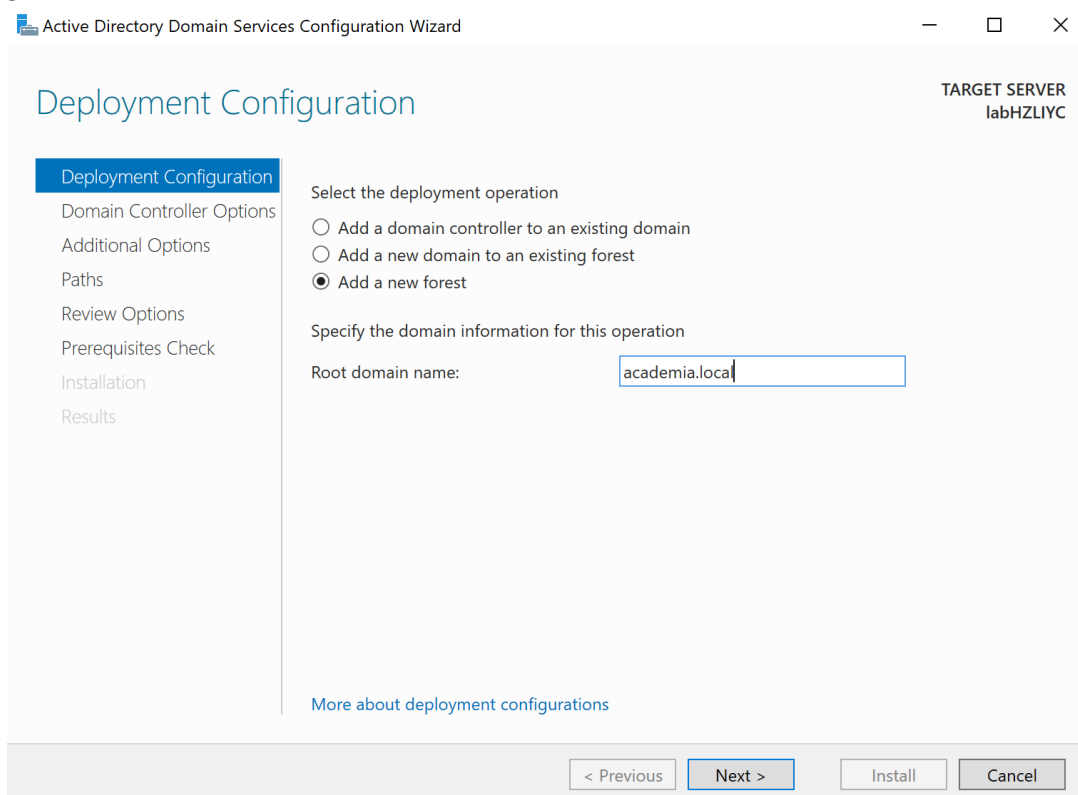
1. Selected AD DS from left navigation pane and got a warning about configuration being required.



2. Clicked 'More' and then clicked on 'Promote server to a domain controller'.



3. Selected a new forest' and entered 'academia.local' for root domain.



4. Provide a DSRM, password as 'Moorhead2026'.

Active Directory Domain Services Configuration Wizard

Domain Controller Options

TARGET SERVER
labHZLIYC

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

Select functional level of the new forest and root domain

Forest functional level: Windows Server 2016

Domain functional level: Windows Server 2016

Specify domain controller capabilities

☒ Domain Name System (DNS) server

☒ Global Catalog (GC)

☐ Read only domain controller (RODC)

Type the Directory Services Restore Mode (DSRM) password

Password:

Confirm password:

[More about domain controller options](#)

< Previous Next > Install Cancel

5. Got the message regarding delegation for DNS server, clicked 'Next' to continue.

Active Directory Domain Services Configuration Wizard

DNS Options

TARGET SERVER
labHZLIYC

Deployment Configuration
Domain Controller Options
DNS Options
Additional Options
Paths
Review Options
Prerequisites Check
Installation
Results

A delegation for this DNS server cannot be created because the authoritative parent zone cannot be found o... [Show more](#) X

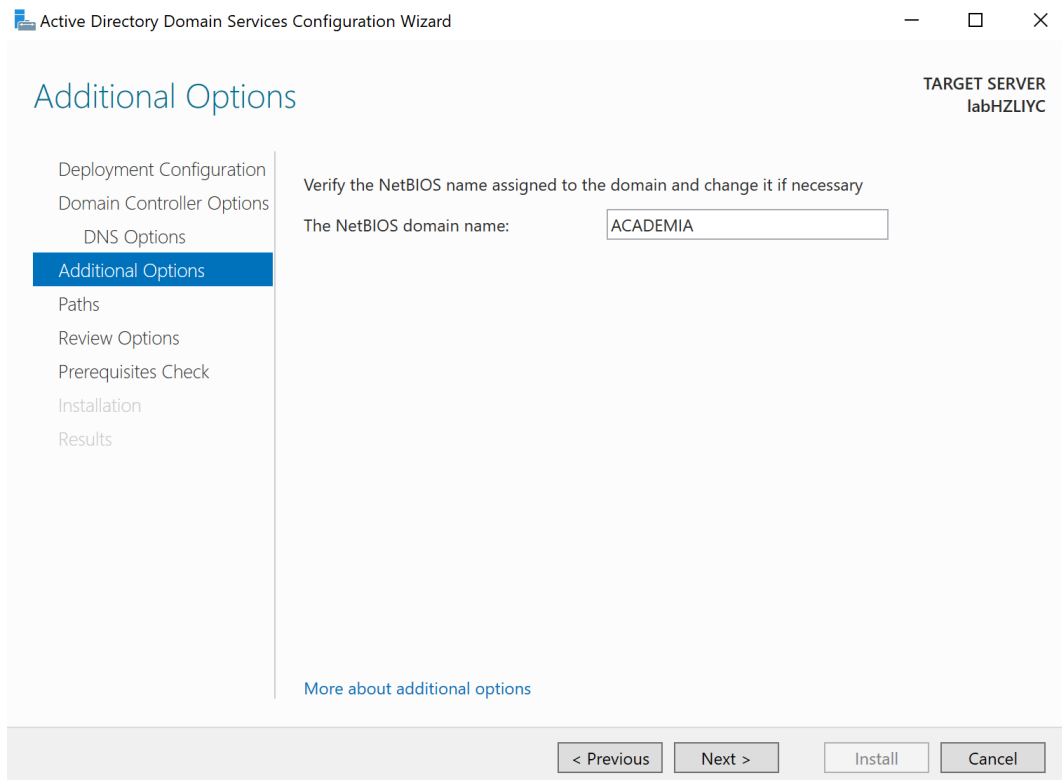
Specify DNS delegation options

☐ Create DNS delegation

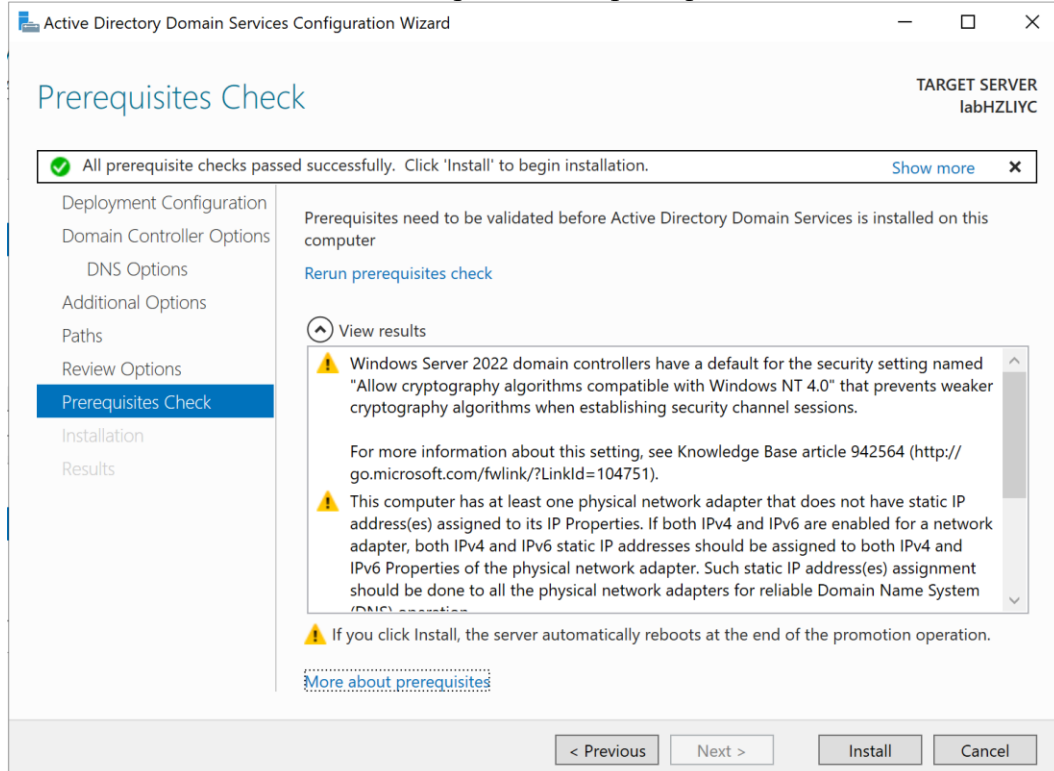
[More about DNS delegation](#)

< Previous Next > Install Cancel

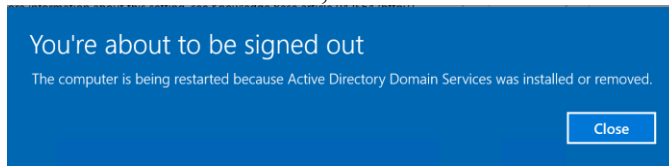
6. When NetBios name is verified, clicked 'Next'.



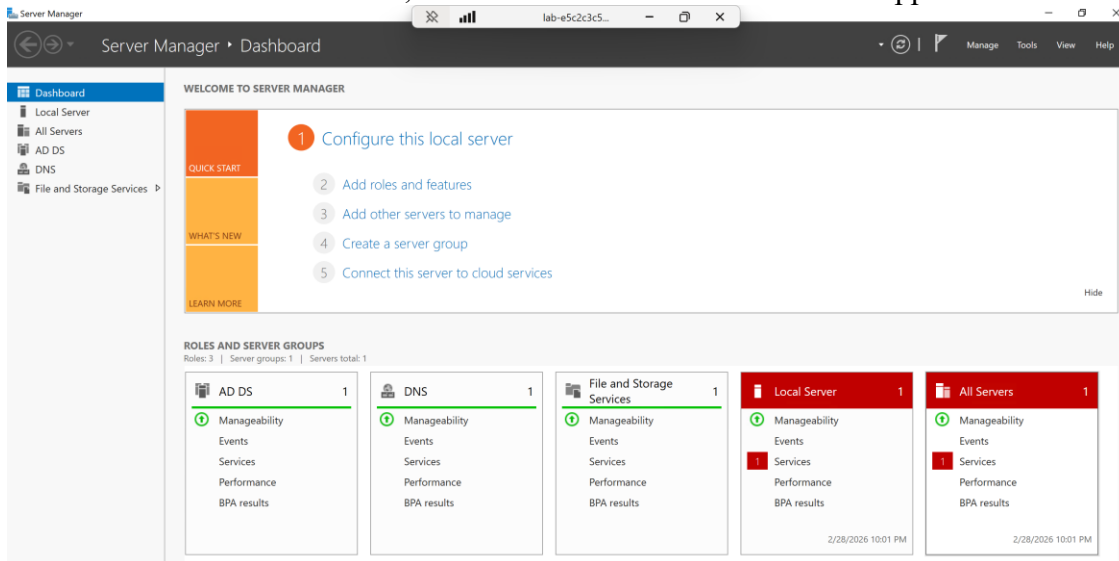
7. Clicked 'Next' for other steps and after prerequisite check, clicked 'Install'.



8. After installation, the Windows Server shutdown.

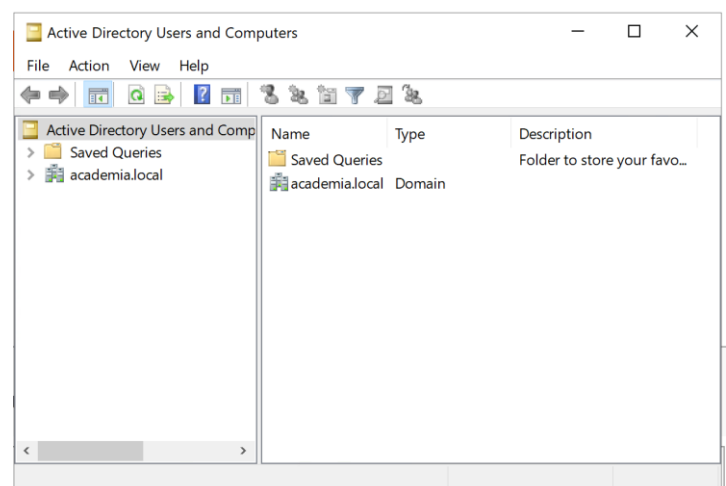
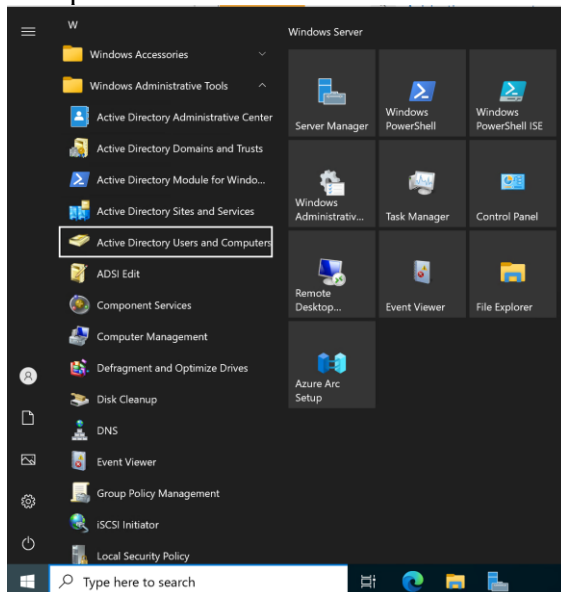


9. After machine restarted, Domain Controller and DNS server appeared on Server Manager.

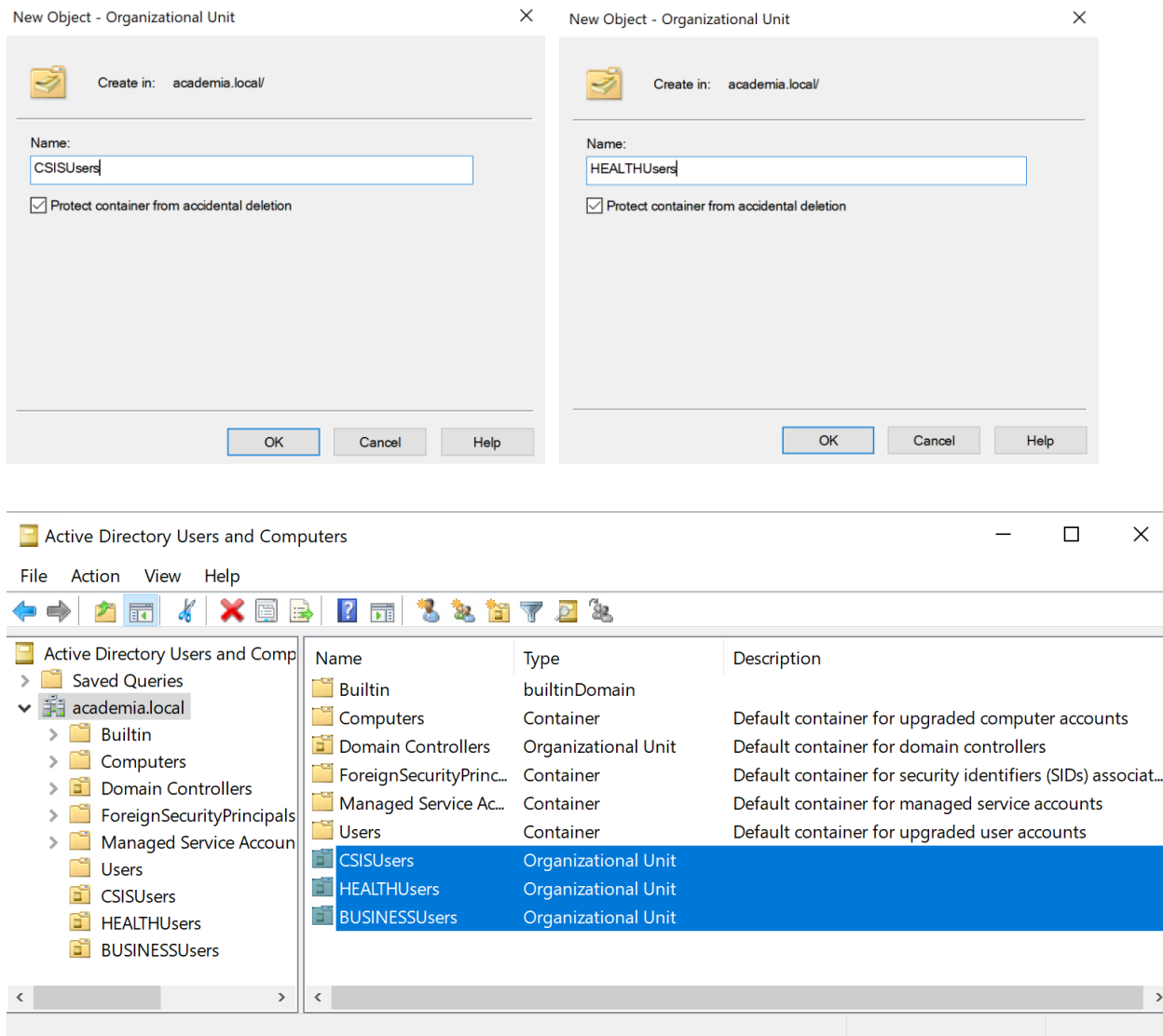


Part 2 – Launching Active Directory Users and Computers

1. Clicked on Windows > Windows Administrative Tools > Active Directory Users and Computers.

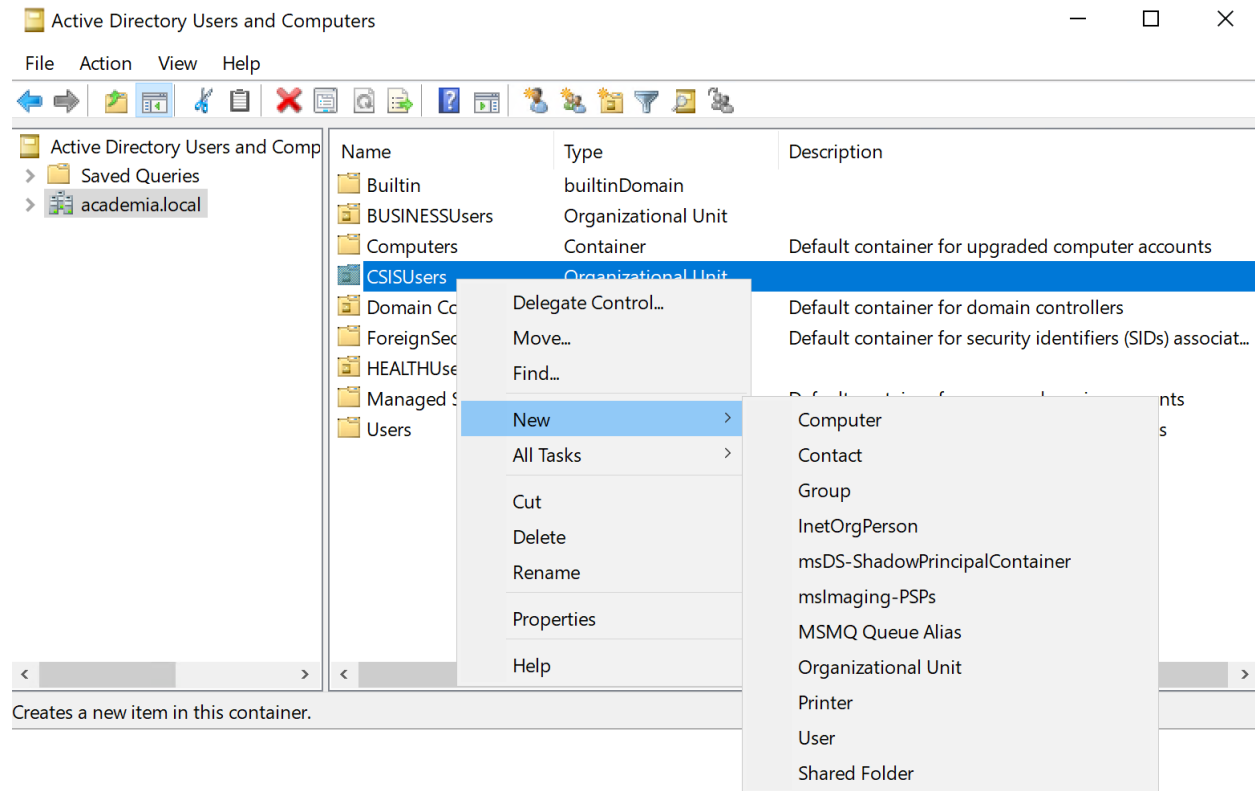


2. Created three Organizational Units (OUs) for CSISUsers, HEALTHUsers, BUSINESSUsers (On the left pane, clicked on academia.local. On the top menu, clicked on 'Action' menu > Select New > Organizational Units).



Part 3 – Creating Users

1. Under academia.local, right-clicked on the CSISUsers OU, select New >User.



2. Provided the user's details as follows.

The 'New Object - User' dialog box is shown. The 'Create in' field is set to 'academia.local/CSISUsers'. The user details are as follows:

Field	Value
First name	Hashan
Last name	Kodippilige
Full name	Hashan Kodippilige
User logon name	HKodippilige@academia.local
User logon name (pre-Windows 2000)	ACADEMIA\HKodippilige

Buttons: < Back, Next >, Cancel

3. Provided password as 'Password01' and checked the box for 'User must change password at next logon'.

New Object - User

Create in: academia.local/CSISUsers

Password:

Confirm:

☒ User must change password at next logon

☐ User cannot change password

☐ Password never expires

☐ Account is disabled

< Back Next > Cancel

New Object - User

Create in: academia.local/CSISUsers

When you click Finish, the following object will be created:

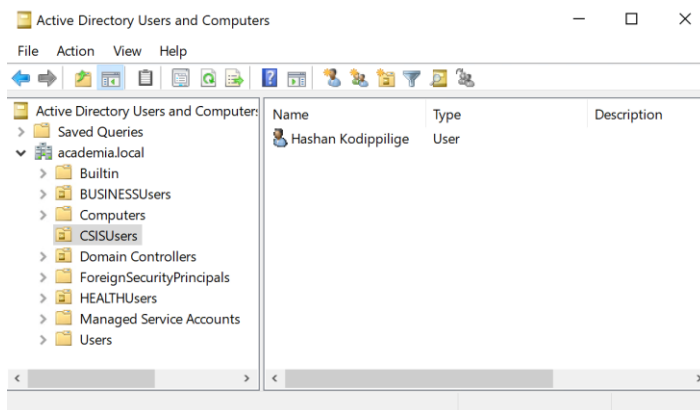
Full name: Hashan Kodippilige

User logon name: HKodippilige@academia.local

The user must change the password at next logon.

< Back Finish Cancel

4. Checked the created user Hashan Kodippilige under CSISUser.



5. Created user Jack Smith under BUSINESSUser (PW- Password01).

New Object - User

Create in: academia.local/BUSINESSUsers

When you click Finish, the following object will be created:

Full name: Jack Smith

User logon name: JSmith@academia.local

The user must change the password at next logon.

< Back Finish Cancel

Active Directory Users and Computers

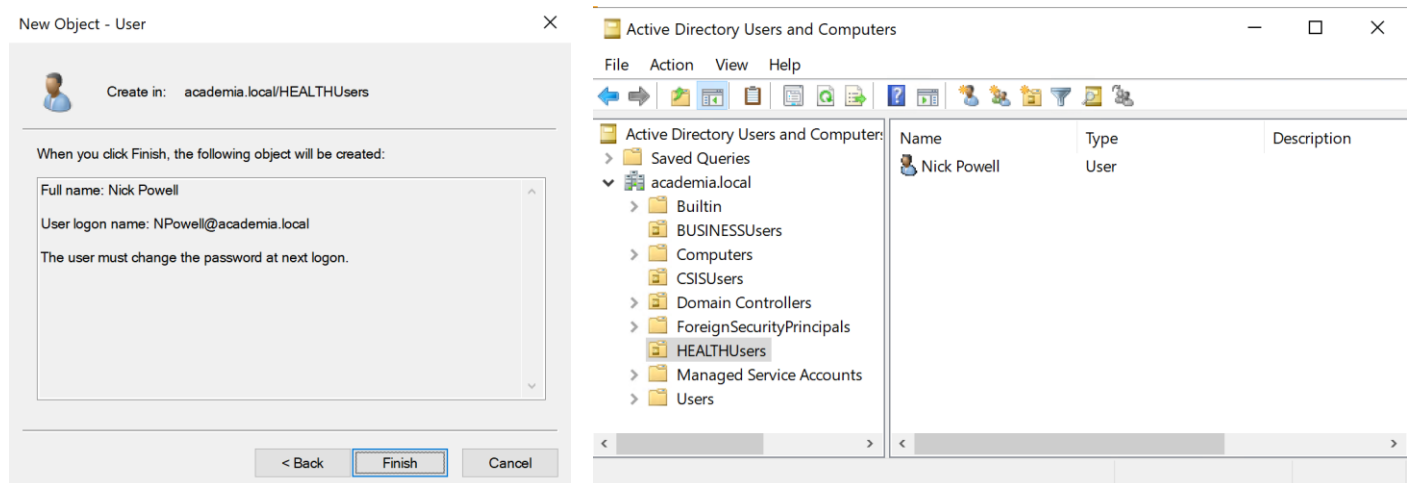
File Action View Help

Active Directory Users and Computers:

- Saved Queries
- academia.local
 - Builtin
 - BUSINESSUsers
 - Computers
 - CSISUsers
 - Domain Controllers
 - ForeignSecurityPrincipals
 - HEALTHUsers
 - Managed Service Accounts
 - Users

Name	Type	Description
Jack Smith	User	

6. Created user Nick Powell under HEALTHUser (PW- Password01).



Part 4 – Questions

1. Explain the use of the Active Directory (AD) tool in cyber operations.

Active Directory plays a vital role in cyber operations because of its centralized control structure. AD manages authentication, authorization, and policy enforcement across an organization. Also, AD verifies user identities and determines what resources they are permitted to access.

Centralized identity management is the primary use of Active Directory in cyber operations. Organizational cybersecurity teams rely on AD to create, modify, disable, and monitor user accounts. Also, when security analysts found out security breach, they can quickly disable compromised accounts to prevent further damage. Using Group Policy Objects (GPOs) which can be created through AD, system administrators can enforce security configurations such as password policies, firewall settings, software restrictions, and account lockout thresholds across domain controllers.

Further, security analysts can detect suspicious activities, failed login attempts, privilege escalation attempts, and lateral movement across the network through security logs generated by domain controllers.

2. Explain cybersecurity risks the Active Directory tool is exposed to.

Active Directory is a powerful administrative tool and a primary target for cyberattacks. Attackers can control over the entire network by compromising Active Directory. The major risk of compromised AD is privilege escalation.

By obtaining access to Active Directory tools, attackers can create accounts, modify policies, and deploy malware across the organization. The credential-based attacks allow attackers to reuse stolen authentication credentials without knowing the actual passwords. Increase the risk of compromising AD tools by having misconfigurations, weak password policies, excessive administrative privileges, and lack of multi-factor authentication. AD is a centralized control structure, and a single successful attack can result in widespread system compromise, data breaches, and operational disruption.

Conclusion

This lab helps to cover the details of Active Directory and demonstrated the installation, configuration, and management of Active Directory Domain Services in a Windows Server 2022 environment. Centralized identity management system was established by promoting the server to a Domain Controller and creating the domain *academia.local*. Allowing to create Organizational units for CSIS, HEALTH, and BUSINESS departments users and creating users for each Organizational units shows how Active Directory allows logical structuring of users and resources within an enterprise. This separation enhances security and credential management.

AD security plays a central role in authentication, authorization, and the overall security of a Windows-based environment (Sophos). Due to these functionalities, it is a high value target for attackers. If the attackers found misconfigured or poorly secured Domain Controller would lead to widespread system access, data breaches, and operational disruption.

This lab provides practical experience of domain structure, OU hierarchy, and user management. Dividing departments highlights the importance of least-privilege access control. Active Directory must be implemented properly and secure to maintain strong cybersecurity defence.

References

1. O'Leary, M. (2019). *Cyber operations: Building, defending, and attacking modern computer networks* (2nd ed.). Apress.
2. Krause, J. (2023). *Mastering Windows Server 2022* (4th ed.). Packt Publishing.
3. Desmond, B., Richards, J., Allen, R., & Lowe-Norris, A. G. (2013). *Active directory* (5th ed.). O'Reilly Media, Inc.
4. Microsoft. (2023). *Active Directory Domain Services overview*. <https://learn.microsoft.com/en-us/windows-server/identity/ad-ds/get-started/virtual-dc/active-directory-domain-services-overview>
5. Sophos. (n.d.). *What is Active Directory in cybersecurity?* <https://www.sophos.com/en-us/cybersecurity-explained/active-directory-security>